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Consider the following Project List and Choose any problem from Set-1 or Set-2:

SET-1

Project Title	Major Modules / Scope
IIPS-MCA Alumni Management System	Alumni Registration, Employment Tracking, Mentorship, Alumni Analytics, Placement Statistics
IIPS UG-PG Academic Project Repository & Record Management System	Project Submission, Repository Management, Guide Allocation, Version Control, Evaluation
Program In-charge Information & Academic Administration System	Faculty Allocation, Course Assignment, Academic Planning, Workload Analysis, Reports
IIPS Laboratory Tracking & Student Performance Analytics System	Laboratory Attendance, Experiment Evaluation, Skill Progress Tracking, CO-PO Mapping Dashboard
IIPS Internship Record Management & Industry Collaboration System	Internship Registration, Company Database, Mentor Evaluation, Offer Tracking, Internship Analytics
Batch Mentor Support & Student Academic Advisory System	Student Profiling, Counseling Records, Attendance Monitoring, Academic Alerts, Mentoring Dashboard
Student Ranking & Outcome-Based Performance Analytics System	Continuous Assessment, CO Attainment, Laboratory Performance, Ranking Algorithms, Academic Analytics
IIPS Sampooran Club Management & Student Activity Information System	Club Registration, Event Planning, Participation Analytics, Leadership Records, Certificate Generation
IIPS Student Poster Submission, Evaluation & Research Ranking System	Poster Submission, Reviewer Assignment, Rubric Evaluation, Research Analytics, Ranking Dashboard



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IIPS Video Assignment Submission, Evaluation & Performance Ranking System	Video Upload, Rubric-based Evaluation, Faculty Review, Peer Assessment, Performance Analytics
Online YouTube Educational Content Analytics System	Channel Analysis, Video Statistics, Engagement Analysis, Learning Recommendation Dashboard
IIPS Digital Learning Resource Repository System	Notes, Presentations, Assignments, Question Bank, Learning Resource Recommendation
IIPS Course Outcome (CO) & Program Outcome (PO) Attainment Analytics System	CO-PO Mapping, Outcome Computation, NBA/NAAC Reports, Accreditation Dashboard
IIPS Academic Performance Prediction & Early Warning System	Attendance Analysis, Internal Assessment Prediction, At-Risk Student Identification
IIPS Research Publication & Faculty Performance Information System	Publications, Patents, Citations, API Score, Faculty Analytics
IIPS Placement Management & Employability Analytics System	Placement Registration, Company Drives, Interview Tracking, Placement Statistics
IIPS Digital Examination & Assessment Management System	Online Tests, Question Bank, Automatic Evaluation, Result Analytics
IIPS Skill Development & Certification Tracking System	NPTEL, SWAYAM, Coursera, Certifications, Student Skill Portfolio
IIPS Innovation, Startup & Incubation Management System	Startup Ideas, Incubation Support, Mentor Allocation, Funding Tracker



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IIPS Research Scholar & Dissertation Management System	Dissertation Workflow, Supervisor Allocation, Progress Monitoring, Plagiarism Reports
IIPS Student Portfolio & Achievement Management System	Academic Portfolio, Technical Skills, Projects, Hackathons, Certifications, Resume Builder
IIPS Smart Classroom Attendance & Learning Analytics System	Attendance, QR/RFID Integration, Participation Tracking, Learning Dashboard
IIPS AI-Based Student Feedback & Sentiment Analysis System	Feedback Collection, NLP-based Sentiment Analysis, Department Quality Dashboard
IIPS Multimodal Student Assignment Analysis, Evaluation & Learning Curve Analytics System	Assignment Submission System : • Text • Image • Audio and • Video Assignments; We have to create AI-assisted Evaluation; (Conventional Method) Faculty and Peer Assessment AI-Assisted System (Proposed) ➤ Similarity Analysis; ➤ Automated Performance Scoring; ➤ Individual & Batch Learning Curve Construction; ➤ Competency Mapping; ➤ Skill Progress Tracking; ➤ Based on Personalized Feedback We have to develop Tool for Learning Improvement Analytics which is capable to asses across 100+ Students;(For multiple Assignment) Dashboard for CO-PO Attainment, Ranking, Trend Analysis and Predictive Performance Monitoring



SET-2

Special Capstone Project (Recommended)

Design and develop the complete Object-Oriented Analysis and Design of an intelligent **Multimodal Student Assignment Analysis and Learning Analytics System** capable of managing, evaluating, and analyzing assignments submitted by **more than 100 students** in multiple digital formats, including **Text, Images, Audio, and Video**.

The proposed system should support assignment submission, rubric-based evaluation, faculty assessment, peer review, automated score computation, competency mapping, plagiarism and similarity reporting (where applicable), feedback generation, and continuous monitoring of individual and batch-wise academic performance.

The system shall analyze historical submission and evaluation data to construct **student learning curves**, identify learning gaps, measure improvement over multiple assessments, compare individual performance with class trends, and generate analytical dashboards for students, mentors, course coordinators, and academic administrators.



TASK

For **Object-Oriented Analysis & Design Laboratory**, every student should select **one** of the above systems and progressively develop:

- Problem Statement
- Stakeholder Analysis
- Software Requirement Specification (SRS)
- Functional & Non-functional Requirements
 - Use Case Model
 - Activity Diagram
 - Class Diagram
 - Object Diagram
 - Sequence Diagram
 - Communication Diagram
 - State Machine Diagram
 - Package Diagram
 - Component Diagram
 - Deployment Diagram
- Database Design (ER Diagram & Relational Schema)
- UI Mockups/Wireframes
- Layered Architecture
- Design Patterns
- Software Design Document (SDD)
- Prototype
- GitHub Repository
- Project Documentation
- Final Presentation